AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 19 · STUDENT LESSON

User Need and AI Contribution

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Before you start

AI-2 Week 32: start with the user's task and a simple baseline.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of capstone requirements and corpus. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of capstone **requirement**s and corpus.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Retrieval Augmented Generation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Verify Numbers and Dates. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Multimodal **Evidence**. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 19: **User**, **Need**, and **AI Contribution**

AI should solve a defined **user** problem better than a **simpler alternative**.



CREATOR MISSION

Compare AI and non-AI designs for a study **need**.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

Recall System evaluation sets. Explain why a recorded result **need**s its **input** conditions and **evidence** source before it can support a claim.

This week's end goal

By the end, I can explain **user need** and **ai contribution** using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied User Need and AI Contribution case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use **user**, **need**, and **ai contribution** to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

A grounded-assistant capstone begins with bounded **requirement**s and a versioned permitted corpus. A baseline can be a manual lookup **process** using the same **evidence**.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

STUDENT LESSON · 3 OF 18

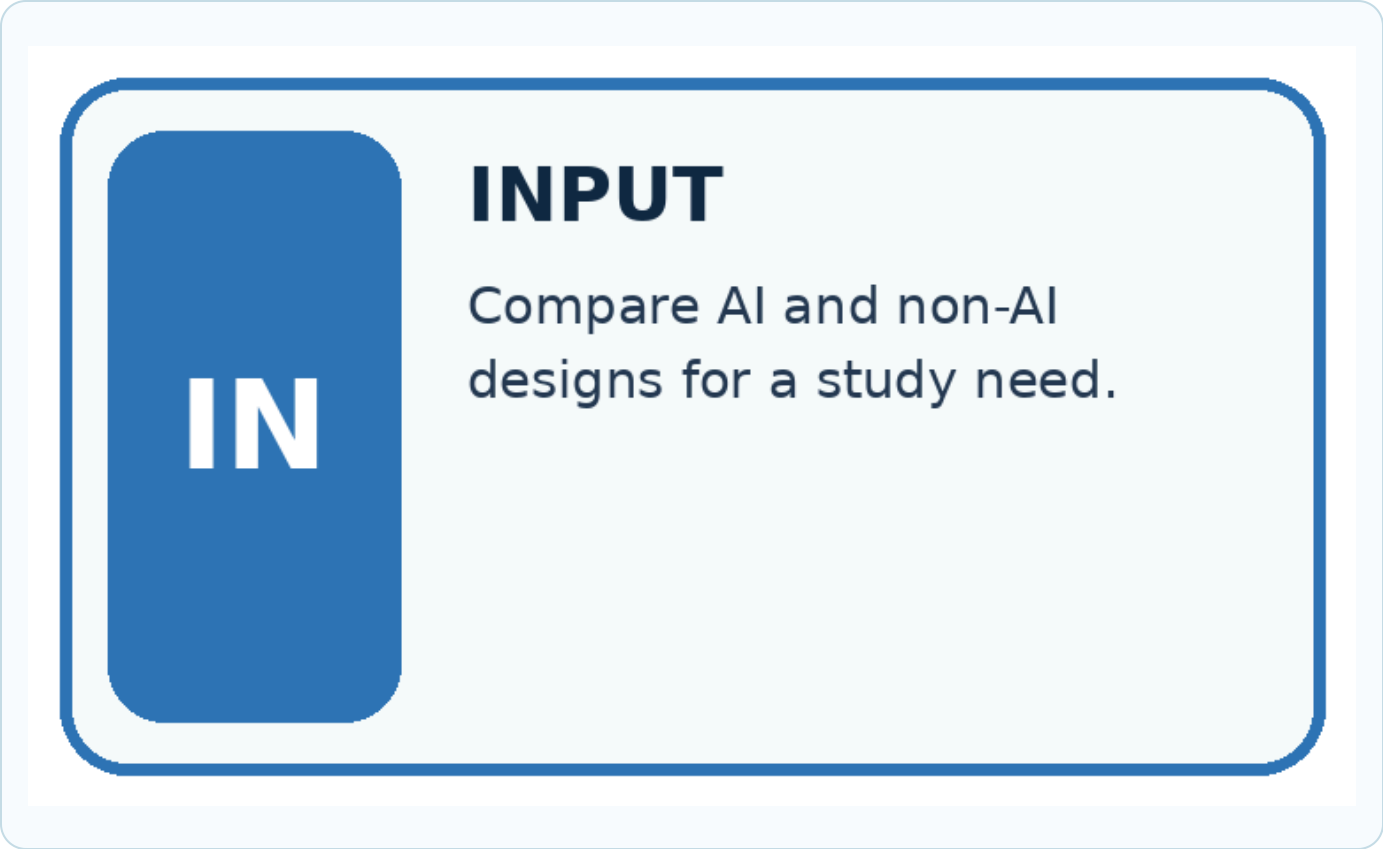
See how it works

> Follow the mechanism for User Need and AI Contribution. Connect each labelled input, deciding step and result.

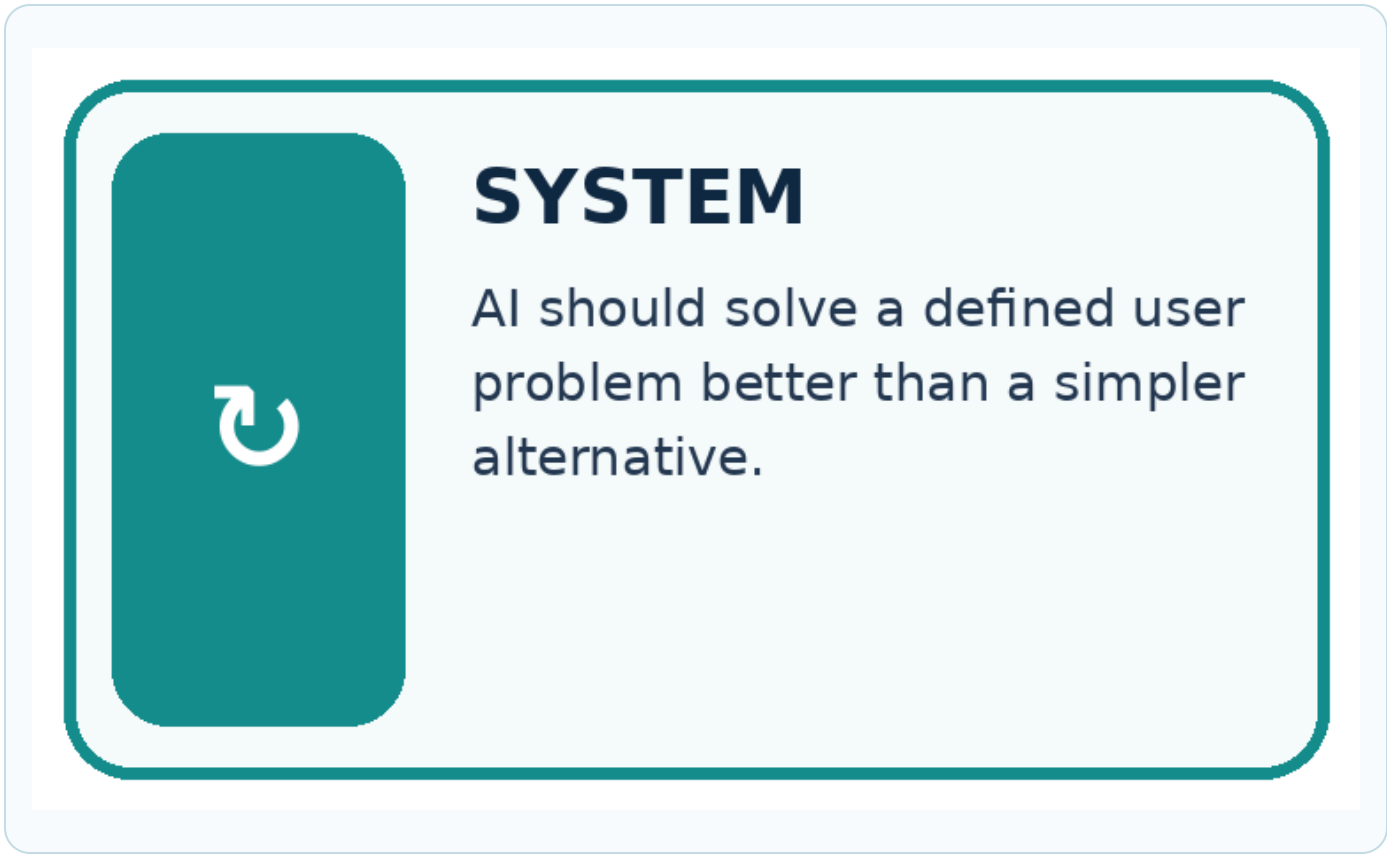
VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓


SYSTEM
AI should solve a defined user problem better than a simpler alternative.
OUT
EVIDENCE
Compare AI and non-AI designs for a study need.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Specify the **user** question, source authority, access, **output**/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation. Keep teacher material out of student retrieval.

Capstone requirements and corpus — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or needs_review.	Specify the user question, source authority, access, output/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation. Keep teacher material out of student retrieval.	Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity. Reserve complete, missing and conflicting cases before development.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

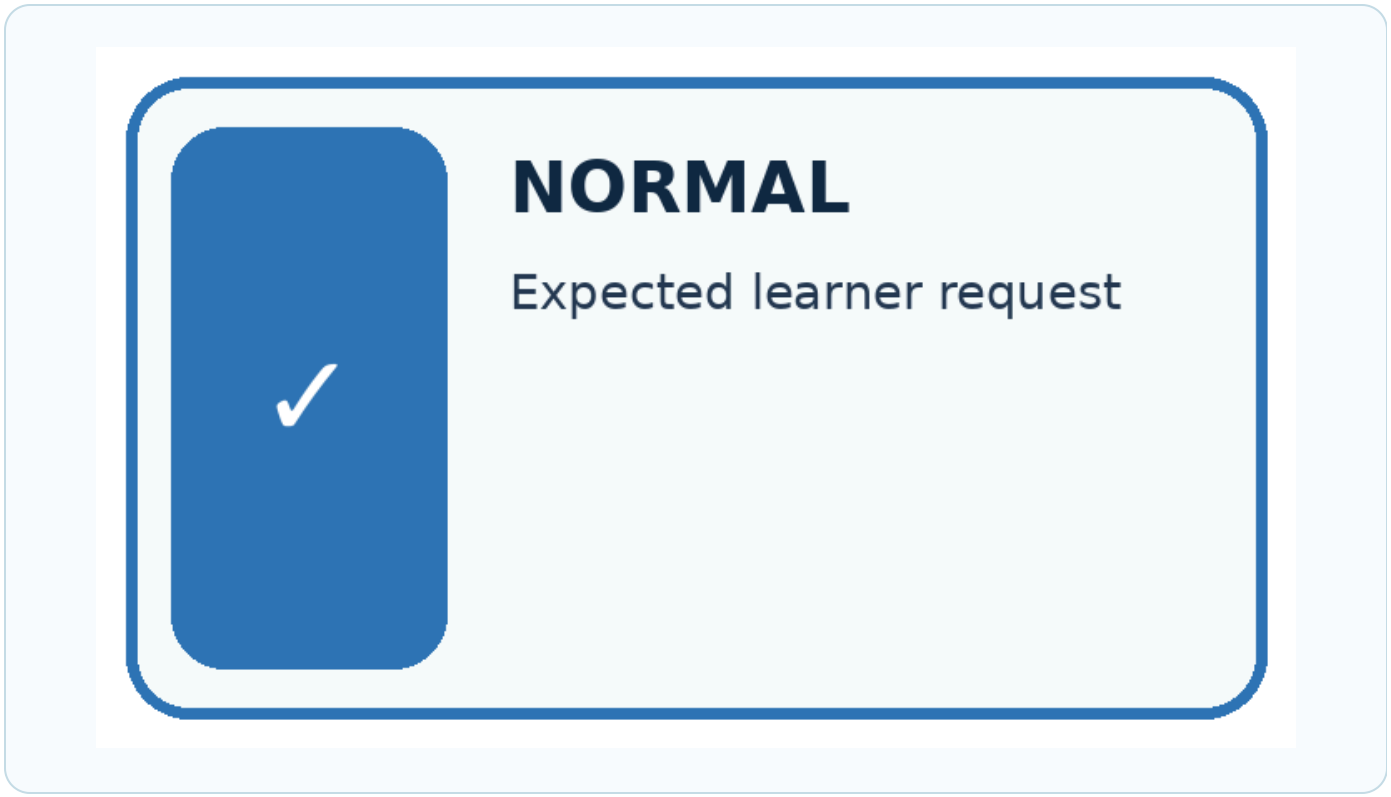
> Compare the supplied cases for User Need and AI Contribution. Identify the detail that changes the conclusion.

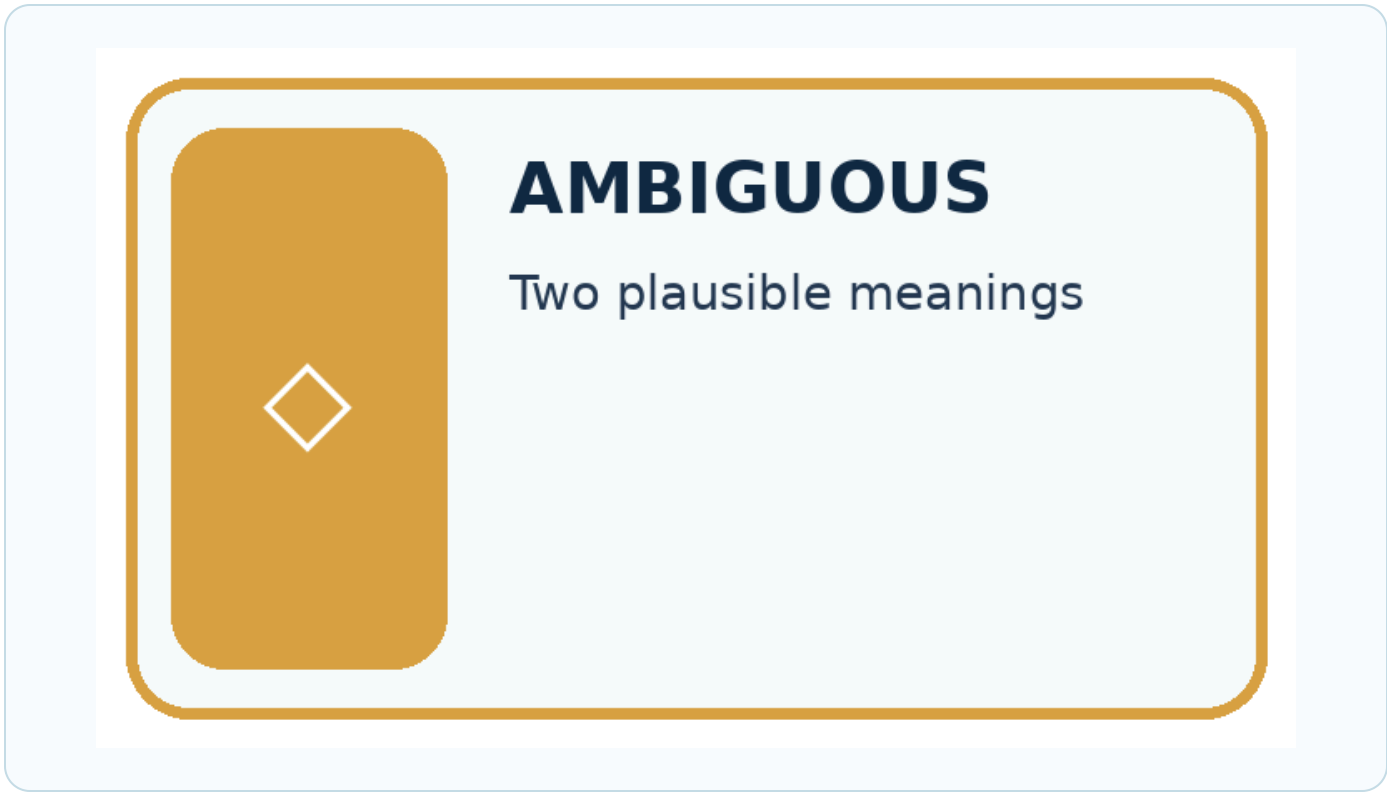
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.


NORMAL
Expected learner request


AMBIGUOUS
Two plausible meanings

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: The pupil asks for photosynthesis, which G3 does not contain.

Predicted result

Prediction: The assistant reports that the definition is unavailable in its approved glossary.

Why the result follows

Reason: A helpful interface cannot expand its authorised **evidence** by guessing.

STUDENT LESSON · 5 OF 18

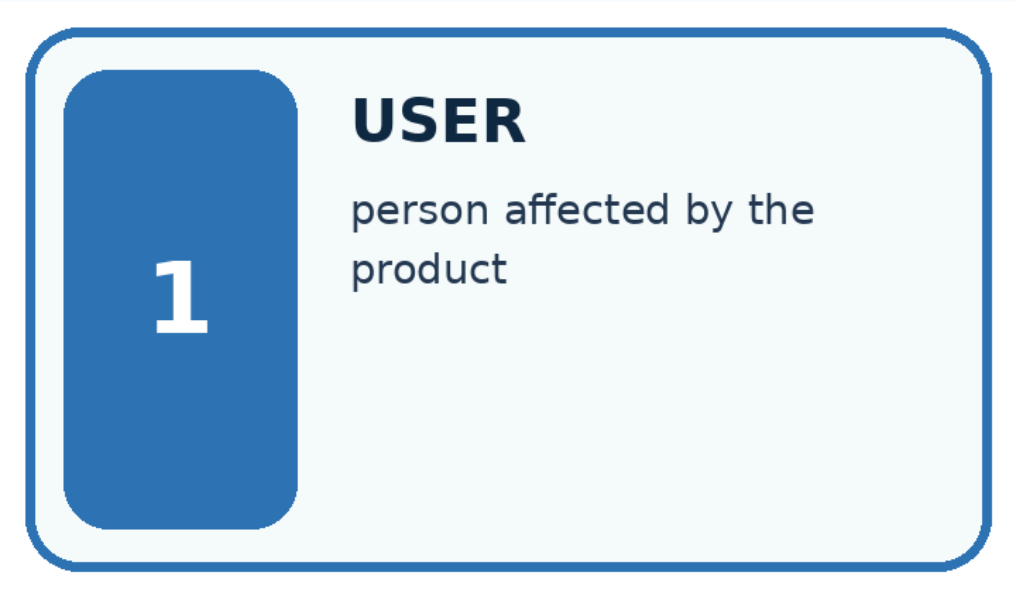
Words that unlock the idea

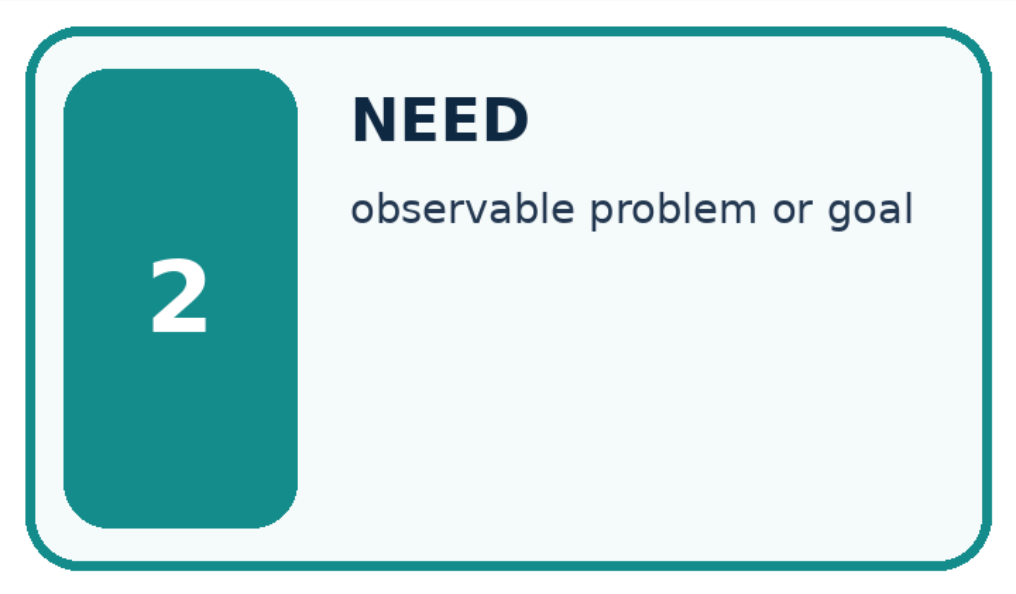
> Use the vocabulary of User Need and AI Contribution to explain a concrete decision, rather than only memorising definitions.

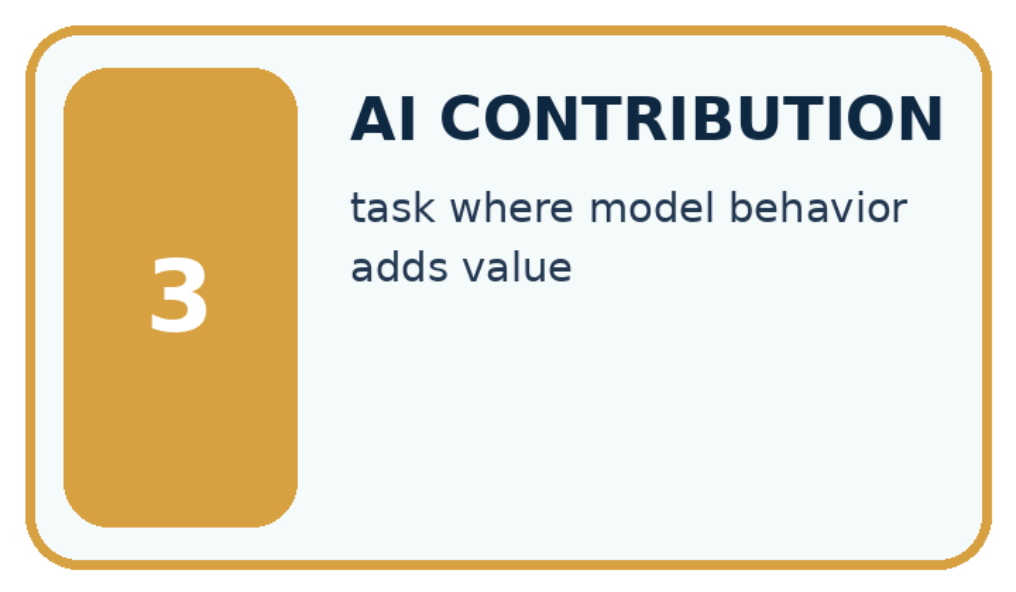
VOCABULARY LAB

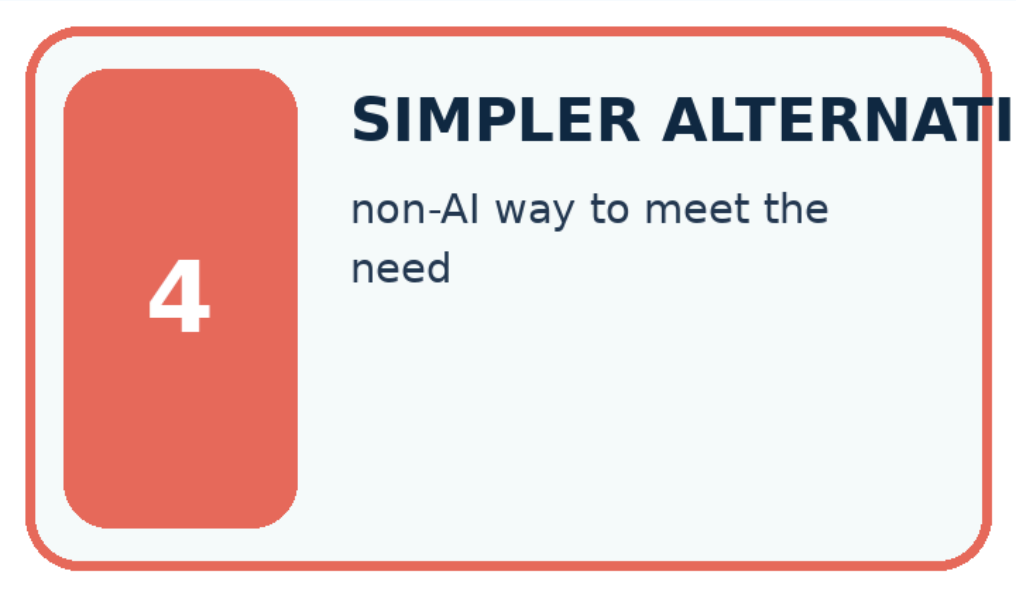
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
USER
person affected by the product


2
NEED
observable problem or goal





Word	My own example	Not this / boundary
user	The person who interacts with or relies on a system.	
need	A problem or desired outcome for the intended user.	
AI contribution	The specific part of a task an AI system is intended to help perform.	

Word	My own example	Not this / boundary
simpler alternative	A less complex approach that may meet the same user need with fewer resources or risks.	

Words used in the closer explanation

Word	Meaning
Requirement	A declared user-visible condition the system must satisfy
Corpus manifest	An inventory of source IDs, versions and access roles
Baseline procedure	A credible reference process on the same task

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STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for User Need and AI Contribution, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 19 • WORKED EVIDENCE

1. Need: turn approved notes into practice questions.
2. Simple template can create fixed question types reliably.
3. AI may add varied wording but also unsupported details.
4. Use AI only where benefit exceeds added verification and safety cost.

1. **Need**: turn approved notes into practice questions.

2. Simple template can create fixed question types reliably.

3. AI may add varied wording but also unsupported details.

4. Use AI only where benefit exceeds added verification and safety cost.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or **need**s_review.

Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity. Reserve complete, missing and conflicting cases before development.

Worked explanation · User Need and AI Contribution

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: User Need and AI Contribution

Define a useful reading helper before choosing its AI features.

The facts we will use

Supplied facts

A pupil **need**s the meaning of unfamiliar words in an approved glossary. G3 defines habitat as the place an organism lives; no other sources are allowed.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/19

User Need and AI Contribution

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name the need

The pupil wants a glossary meaning, not a made-up example presented as a fact.



2. Choose contribution

The assistant locates the glossary entry and restates it in clear language with G3.



3. Bound the service

An absent word produces "not in this glossary"; a direct glossary search is the baseline.

CHANGE ONE THING

The pupil asks for photosynthesis, which G3 does not contain.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Name the need	The pupil wants a glossary meaning, not a made-up example presented as a fact.
2. Choose contribution	The assistant locates the glossary entry and restates it in clear language with G3.
3. Bound the service	An absent word produces “not in this glossary”; a direct glossary search is the baseline.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. An absent word produces “not in this glossary”; a direct glossary search is the baseline. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

A **requirement**s plan identifies intended value; it does not demonstrate actual pupil benefit.

Keep the stated limit

Read the labelled picture: What non-AI baseline would reveal whether this helper adds value? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: The pupil asks for photosynthesis, which G3 does not contain. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the User Need and AI Contribution claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Adding AI automatically makes a product more useful. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

STUDENT LESSON · 8 OF 18

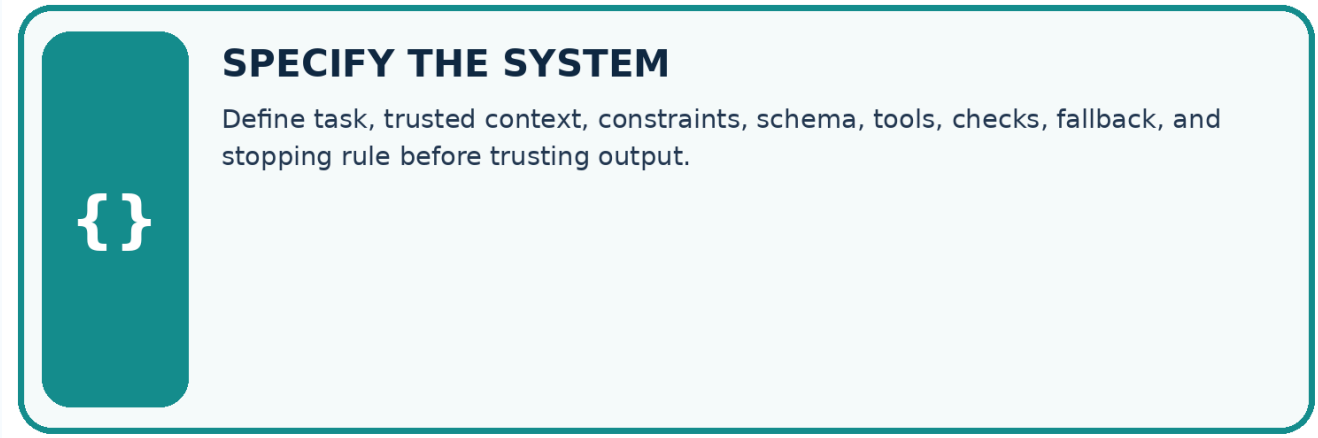
Practise together

Practise User Need and AI Contribution with support, then explain the deciding step in your own words.

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

A complete capstone plan is not **evidence** that the implementation meets it; evaluate after building.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Capstone **requirement**s and corpus”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Capstone **requirement**s and corpus. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if **need**ed.

Answer on your own before discussion. If you are unsure, say which step you **need** help with.

Explain the picture

What non-AI baseline would reveal whether this helper adds value? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

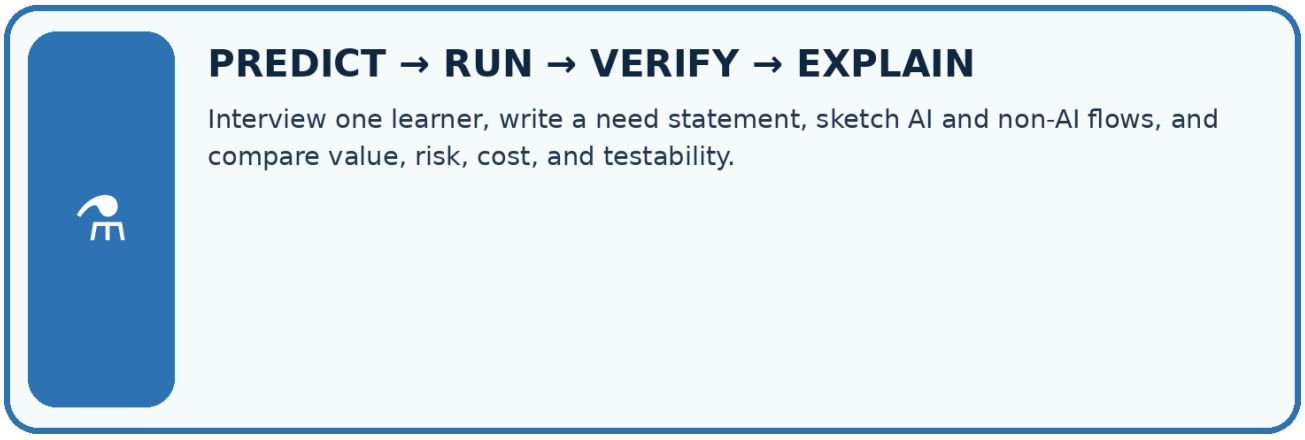
Run the investigation

> Investigate User Need and AI Contribution. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Interview one learner, write a **need** statement, sketch AI and non-AI flows, and compare value, risk, cost, and testability.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

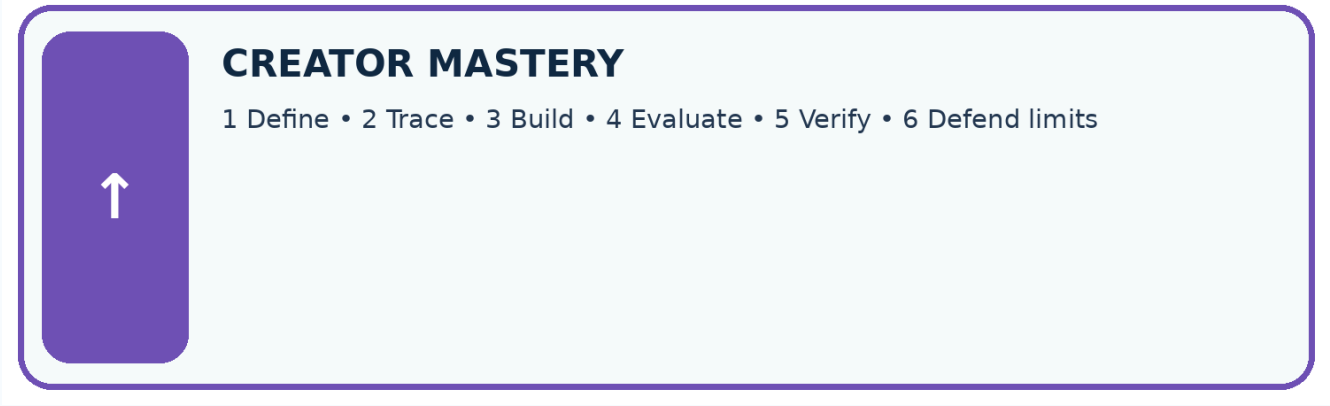
Build your understanding

> Move from recall to tracing, diagnosis and transfer in User Need and AI Contribution. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

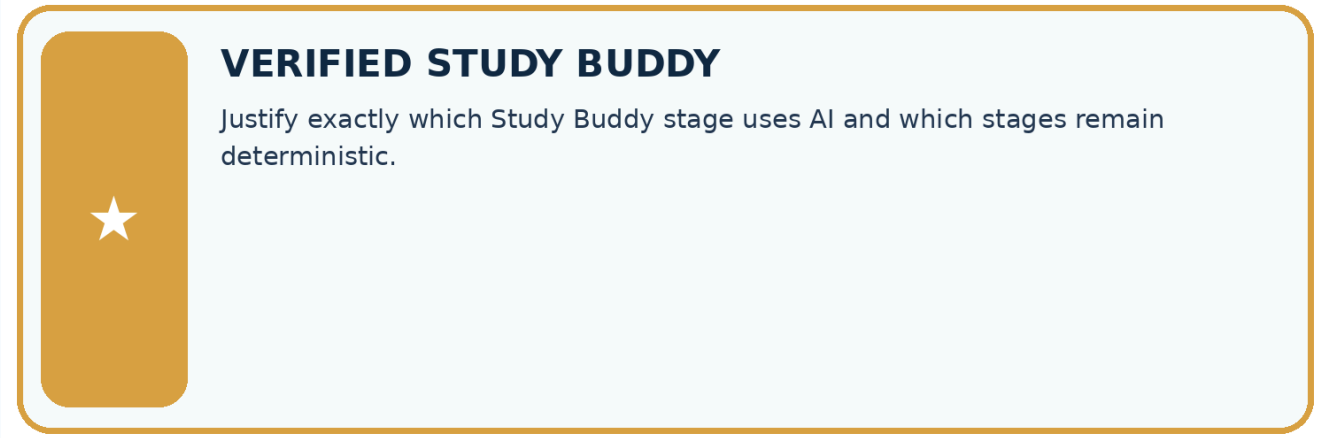
Connect your project

> Apply this week's idea to Verified Study Buddy. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Justify exactly which Study Buddy stage uses AI and which stages remain deterministic.

Trace the system

SYSTEM MAP Learner **need** → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

Evidence for your project

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

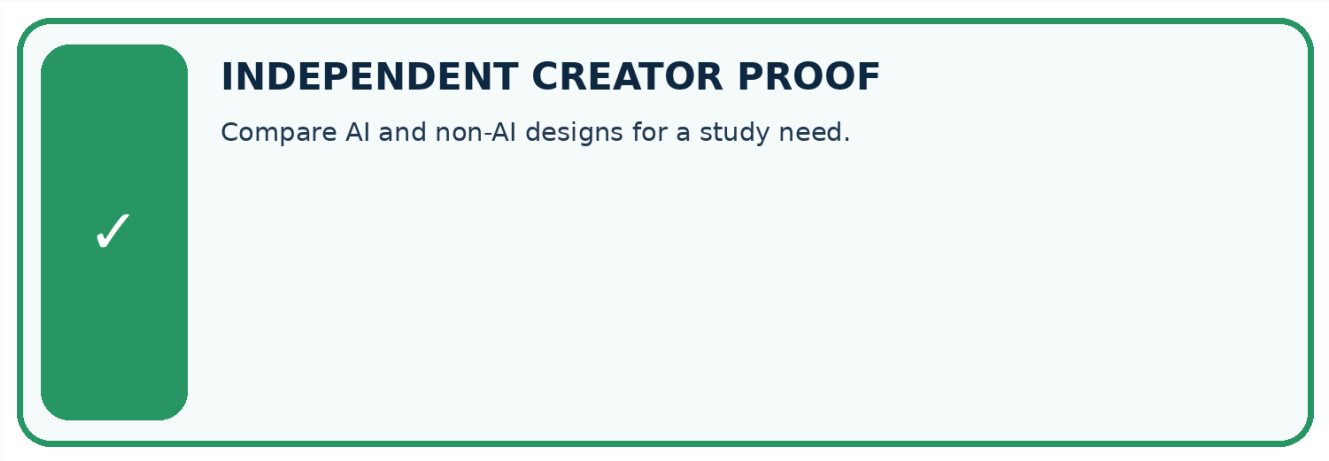
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Compare AI and non-AI designs for a study **need**.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation **need**ing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete User Need and AI Contribution case for supported practice. Repeated examples do not become fresh assessments.

Week 19 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Learners struggle to locate definitions in an approved chapter; a proposed chatbot answers everything.

The question

Narrow the **need** and **AI contribution**.

Worked explanation

Help locate and explain chapter definitions with references and unsupported-question handling. Compare a simple index or search first.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Choose a narrow, testable contribution

Evidence or stage	Value or result
Need	Find chapter definitions
Simple baseline	Index or search

Evidence or stage	Value or result
AI help	Explain with chapter references
Limit	Handle unsupported questions

Learners struggle to locate definitions in an approved chapter; a proposed chatbot answers everything.

Narrow the **need** and **AI contribution**.

Help locate and explain chapter definitions with references and unsupported-question handling. Compare a simple index or search first.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Student equipment assistant has public inventory I, restricted purchase ledger L and public booking notice B. Define eligible corpus, baseline and a missing-**evidence** test.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 19 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A student **need**s the exact due date from a course notice. Should the design start by generating a plausible date or locating the notice? Explain the **AI contribution**.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Justify exactly which Study Buddy stage uses AI and which stages remain deterministic.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Students cannot find definitions in a 20-page handout. Compare an index with a grounded assistant. Define a success measure and a failure case for both.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate User Need and AI Contribution, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Define a useful reading helper before choosing its AI features.

By the end, I can	How I will show it
Explain the mechanism and terminology of capstone requirements and corpus	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain User Need and AI Contribution without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of capstone **requirement**s and corpus

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Define a useful reading helper before choosing its AI features.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? An absent word produces “not in this glossary”; a direct glossary search is the baseline.

1. Bound the service

2. Name the **need**

3. Choose contribution

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if **need**ed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Bound the service An absent word produces “not in this glossary”; a direct glossary search is the baseline. Read the picture in order.

An absent word produces “not in this glossary”; a direct glossary search is the baseline. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The pupil asks for photosynthesis, which G3 does not contain. Which conclusion is supported by the supplied facts?

1. The assistant reports that the definition is unavailable in its approved glossary.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if **need**ed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The assistant reports that the definition is unavailable in its approved glossary. A helpful interface cannot expand its authorised **evidence** by guessing.

Repair a missing building block

Review Retrieval Augmented Generation (ai-3/18). Goal: Explain the **mechanism** and terminology of building a grounded answer. Short practice: Which part of the source supports a two-minute melting claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Verify Numbers and Dates (ai-3/16). Goal: Explain the **mechanism** and terminology of claim support and citation evaluation.

Short practice: Which part can be corrected immediately from existing **evidence**, and which **need**s follow-up?. Return to this week and explain what you changed.

Review Multimodal **Evidence** (ai-3/13). Goal: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**.

Short practice: Why were both “exactly three” and “exactly five” unsupported before opening the box?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

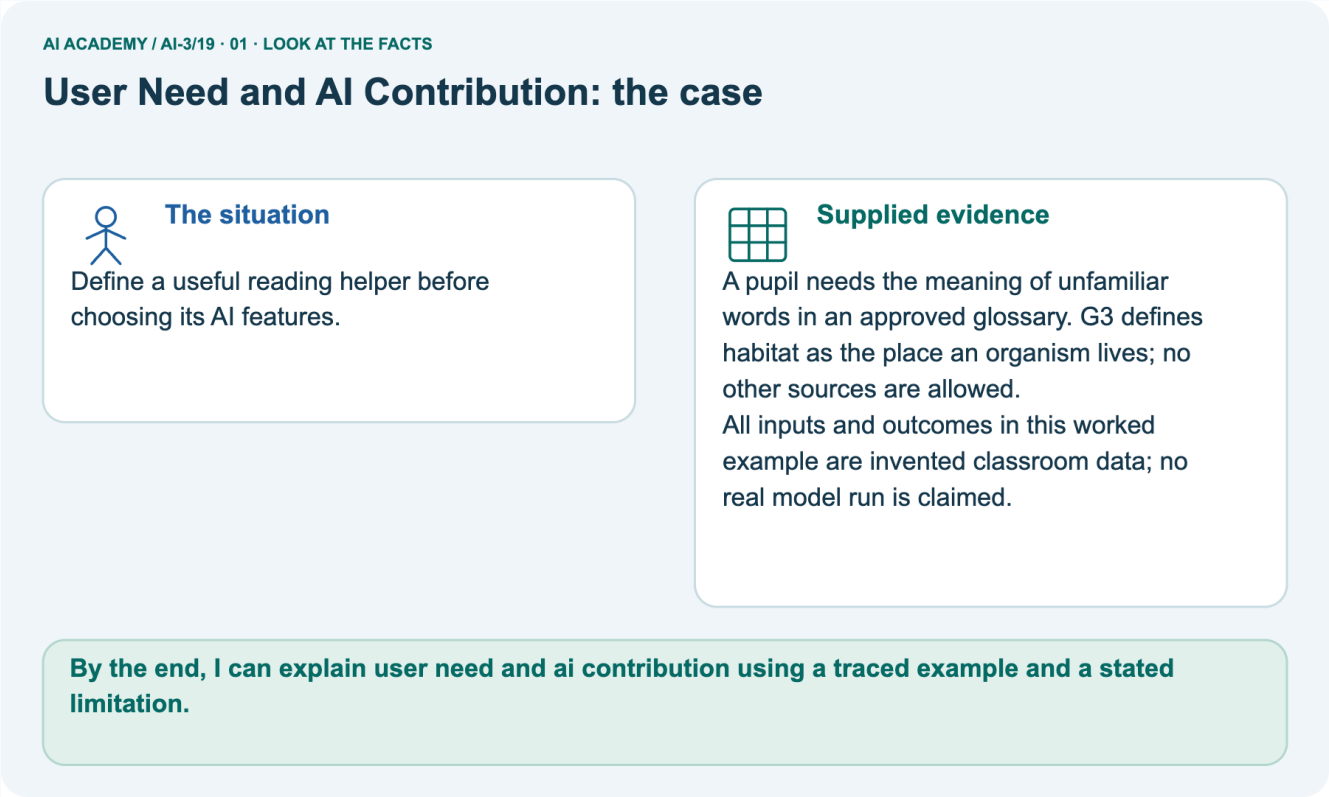
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each User Need and AI Contribution graphic using its labels, deciding evidence and relevant limitation.

01. User Need and AI Contribution: the case



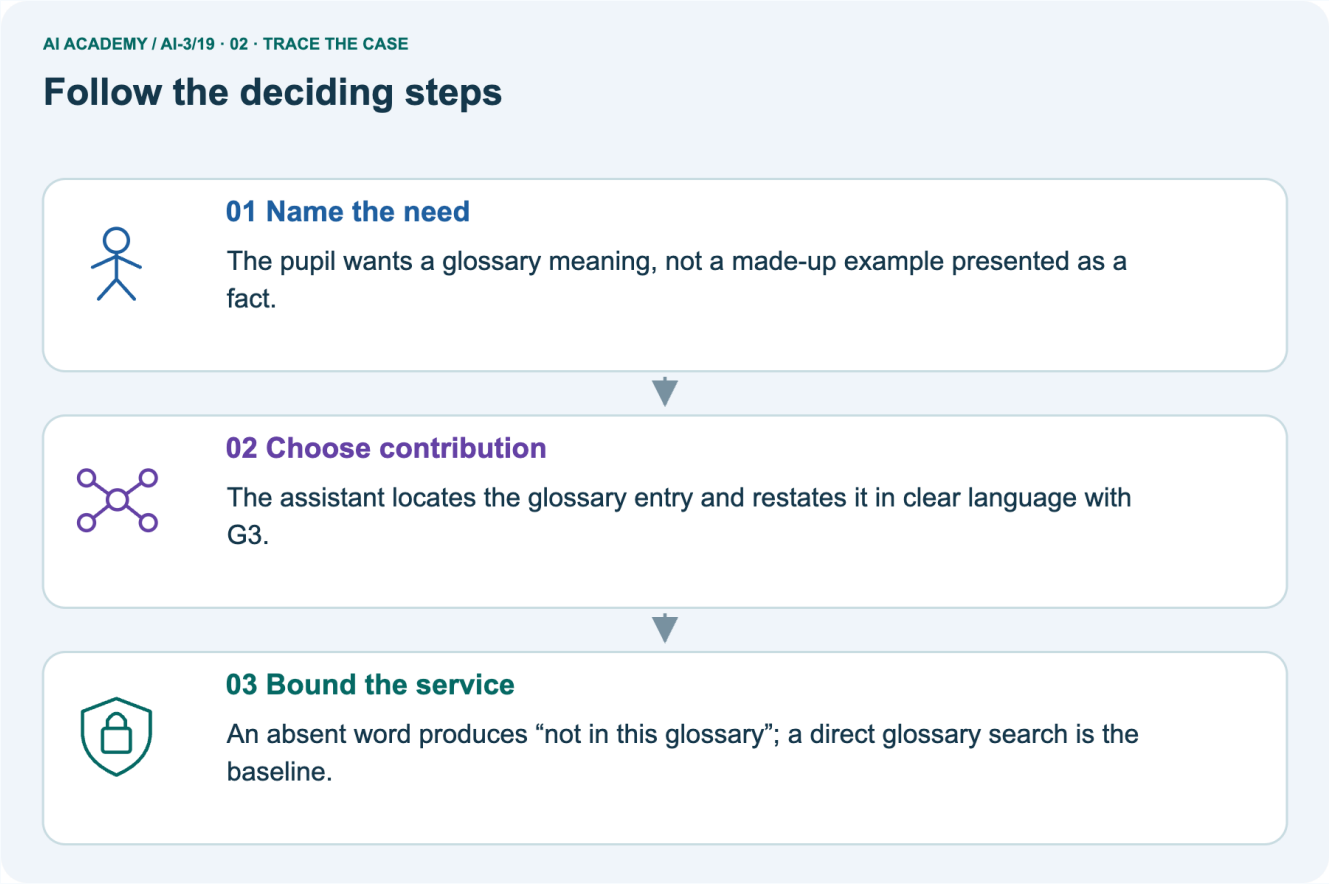
Follow the labelled picture with the matching explanation. Open the vector diagram

Define a useful reading helper before choosing its AI features. Supplied facts: A pupil **need**s the meaning of unfamiliar words in an approved glossary.

G3 defines habitat as the place an organism lives; no other sources are allowed. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

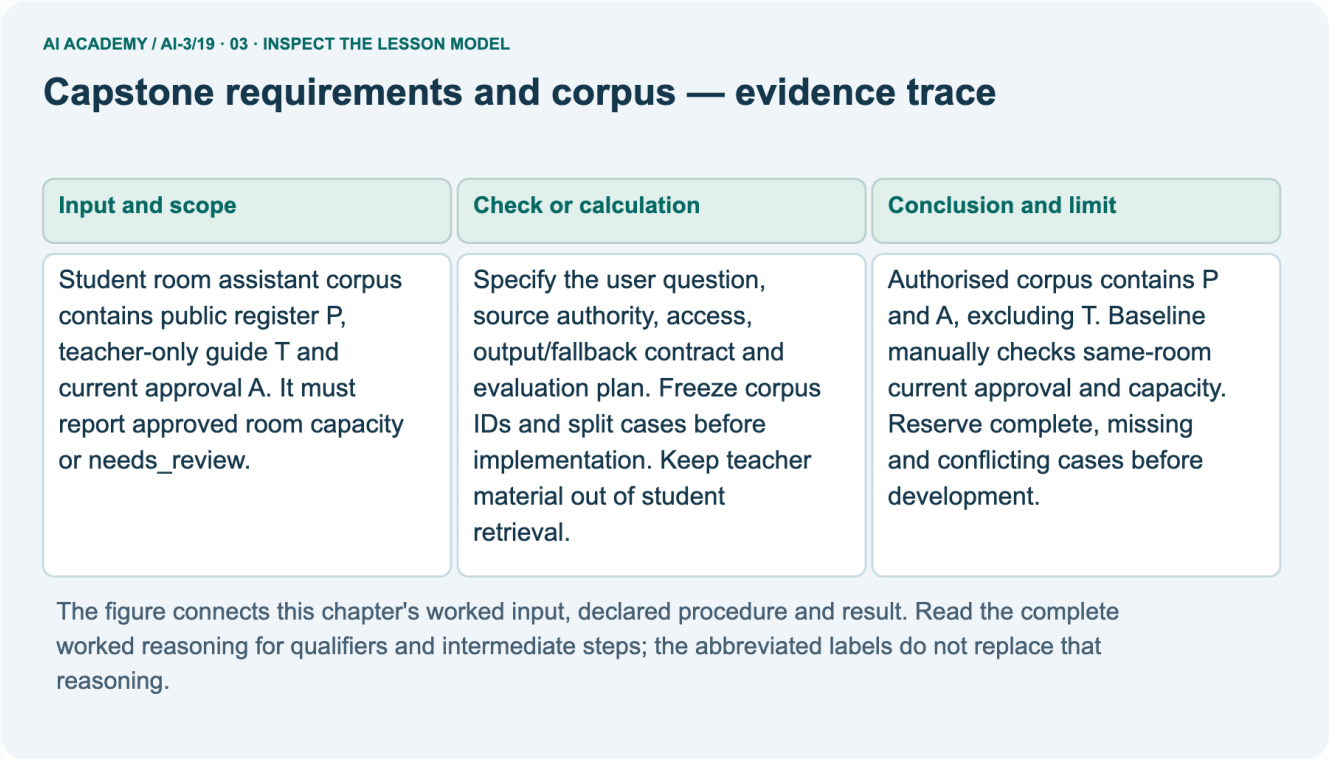


Follow the labelled picture with the matching explanation. Open the vector diagram

Name the **need**: The pupil wants a glossary meaning, not a made-up example presented as a fact. → Choose contribution: The assistant locates the glossary entry and restates it in clear language with G3. → Bound the service: An absent word produces “not in this glossary”; a direct glossary search is the baseline.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Capstone requirements and corpus — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone **requirement**s and corpus — **evidence** trace. Student room assistant corpus contains public register P, teacher-only guide T and current approval A.

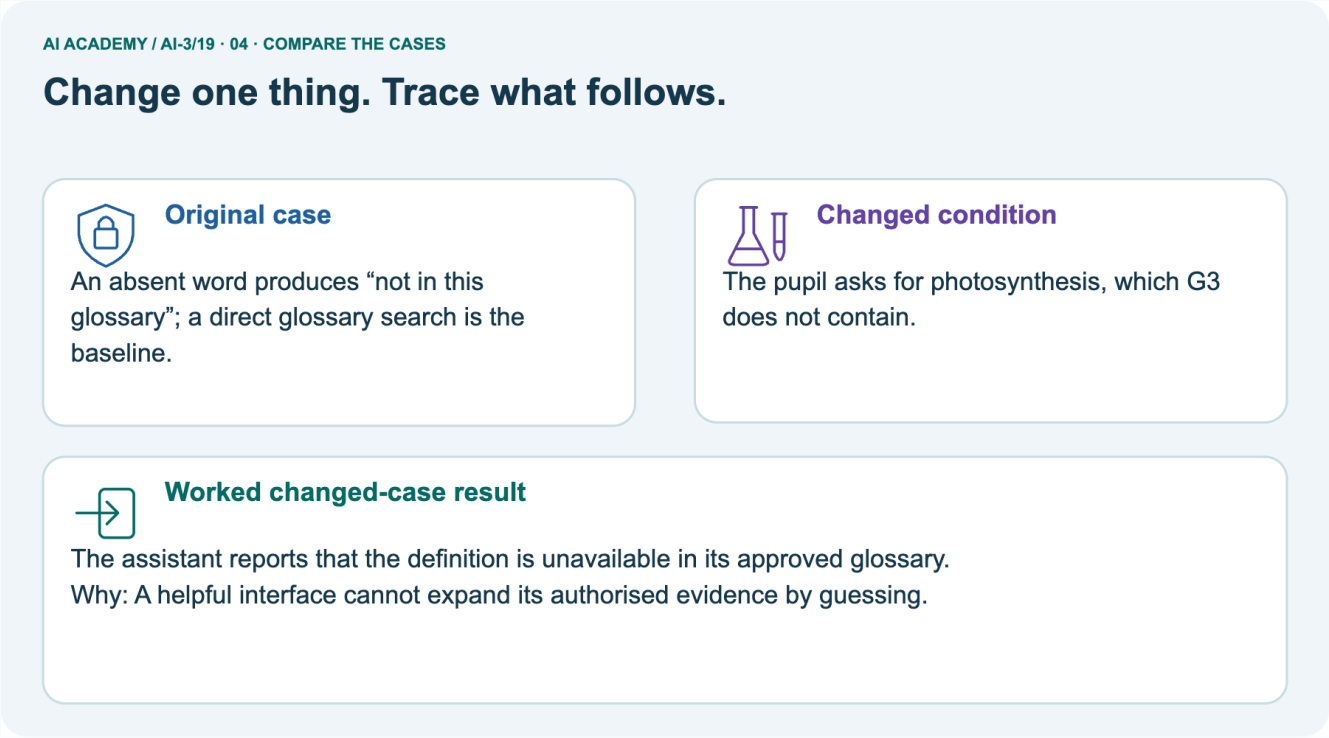
It must report approved room capacity or **need**s_review. | Specify the **user** question, source authority, access, **output**/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation.

Keep teacher material out of student retrieval. | Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity.

Reserve complete, missing and conflicting cases before development.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



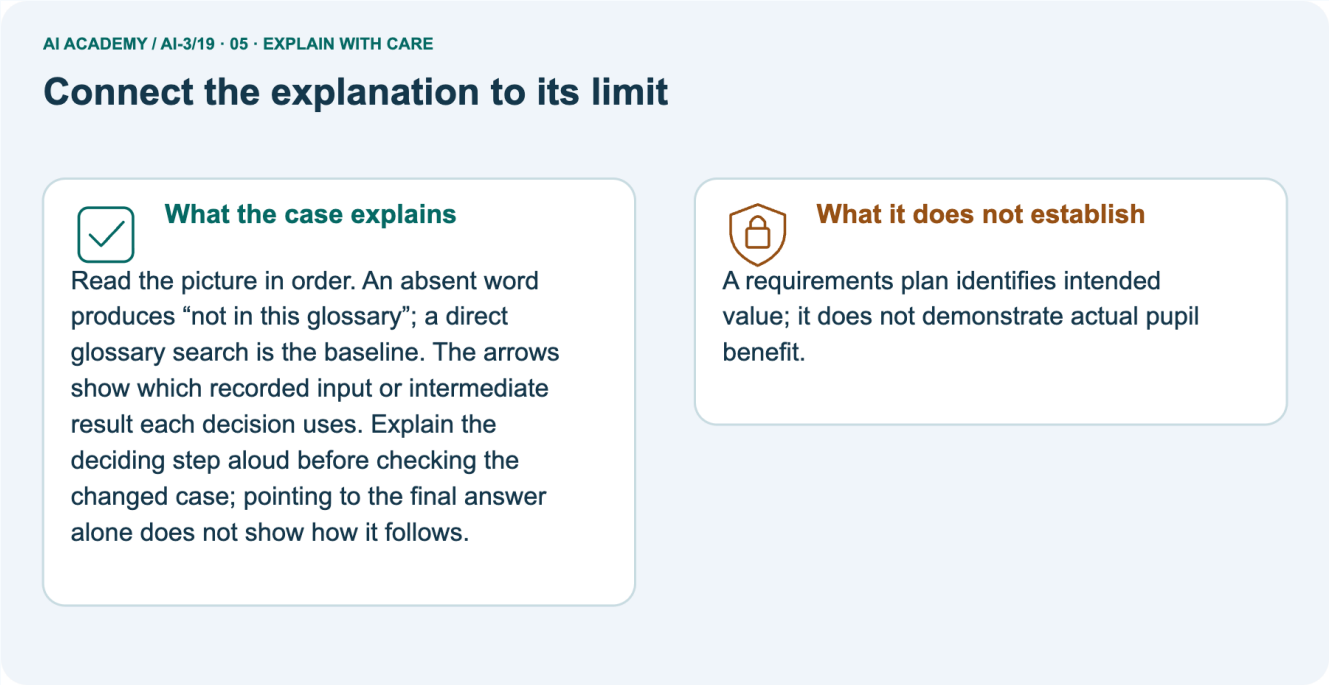
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: An absent word produces “not in this glossary”; a direct glossary search is the baseline. Changed condition: The pupil asks for photosynthesis, which G3 does not contain.

Expected result: The assistant reports that the definition is unavailable in its approved glossary. Reason: A helpful interface cannot expand its authorised **evidence** by guessing.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

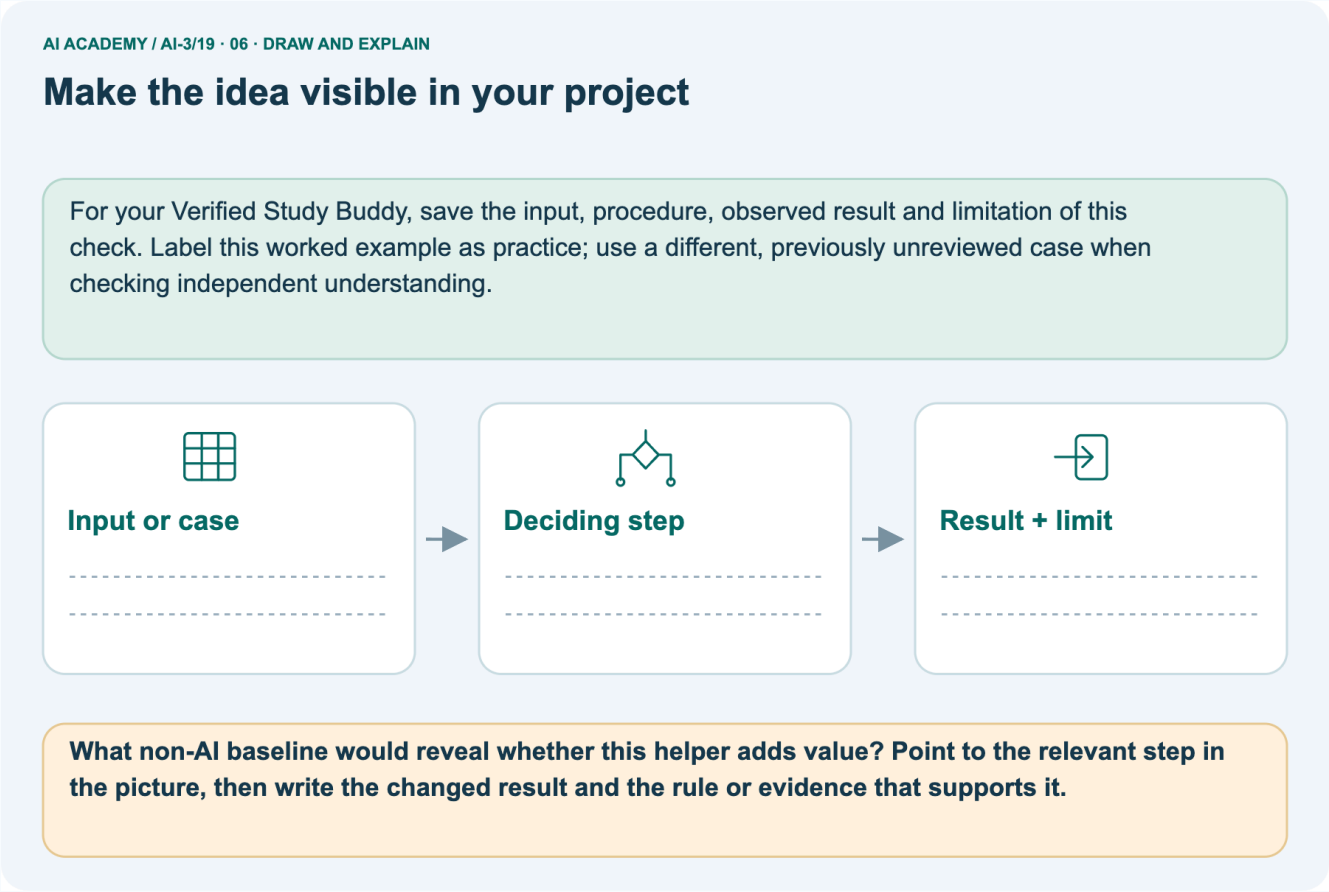
Read the picture in order. An absent word produces “not in this glossary”; a direct glossary search is the baseline. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A **requirement**s plan identifies intended value; it does not demonstrate actual pupil benefit.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What non-AI baseline would reveal whether this helper adds value? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.